A streaming brain-to-voice neuroprosthesis to restore naturalistic communication

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Abstract

Natural spoken communication happens instantaneously. Speech delays longer than a few seconds can disrupt the natural flow of conversation. This makes it difficult for individuals with paralysis to participate in meaningful dialogue, potentially leading to feelings of isolation and frustration. Here we used high-density surface recordings of the speech sensorimotor cortex in a clinical trial participant with severe paralysis and anarthria to drive a continuously streaming naturalistic speech synthesizer. We designed and used deep learning recurrent neural network transducer models to achieve online large-vocabulary intelligible fluent speech synthesis personalized to the participant's preinjury voice with neural decoding in 80-ms increments. Offline, the models demonstrated implicit speech detection capabilities and could continuously decode speech indefinitely, enabling uninterrupted use of the decoder and further increasing speed. Our framework also successfully generalized to other silent-speech interfaces, including single-unit recordings and electromyography. Our findings introduce a speech-neuroprosthetic paradigm to restore naturalistic spoken communication to people with paralysis.